



FIREBRAND

LYCEUM EDUCATION GROUP

Data Structures and Storage in Power Platform

Power Platform Pathway | Session 4

Session Recording & Professional Expectations

Session Recording

- All Estio training sessions are recorded to:
 - Support learner progress
 - Enable review of content
 - Maintain quality and compliance standards
- By attending this session, you acknowledge that recording is taking place.

Camera & Participation Expectations

- To support engagement and attendance monitoring:
 - Cameras should remain on during the session
 - Please participate actively in discussions and activities
 - If you are unable to use your camera or microphone for any reason, please inform your tutor directly via chat or message.

This session forms part of your professional apprenticeship experience. Please engage respectfully and professionally throughout.

Safeguarding, British Values & Wider Learning

Safeguarding protects individuals from harm, abuse, neglect, radicalisation and exploitation.

If you have a concern about yourself or someone else, email:

safeguarding@bpp.com

Or speak to your tutor privately.

We treat all concerns seriously and confidentially.

Safeguarding



The British Values are:

- 1. Democracy**
- 2. Rule of Law**
- 3. Individual Liberty**
- 4. Mutual Respect**
- 5. Tolerance of Different Faiths and Beliefs**

These values underpin everything we do in professional and digital environments.

British Values



Prevent is about protecting people from being drawn into extremism or radicalisation.

In our programmes, Prevent links to:

- **Online safety**
- **Digital responsibility**
- **Critical thinking**
- **Professional ethics**

Prevent



As we go through today's session, consider:

How today's topic connects to British Values

Whether there are any Prevent or digital safety themes

How this learning supports your professional and career development

Your tutor will guide discussion and help you make these connections.

Competencies and Coverage

Professional Discussion underpinned by Portfolio

Knowledge

K1, K3, K4, K5, **K7**, K8,
K10, K12

Skills

S2, S3, S5, S8, S9, S13,
S14, S15, S17

Behaviours

B1, B4, B5, B6, B7, B8,
B9

Work-based Project with Questioning

Knowledge

K2, K6, **K9**, K11

Skills

S1, S4, S6, S7, S10,
S11, S12, **S16**

Behaviours

B2, B3

Lesson Aim and Objectives

Lesson Aim: Data Structures and Storage in Power Platform

Lesson Objectives

Apply

- Apply structured data storage in apps

Analyse

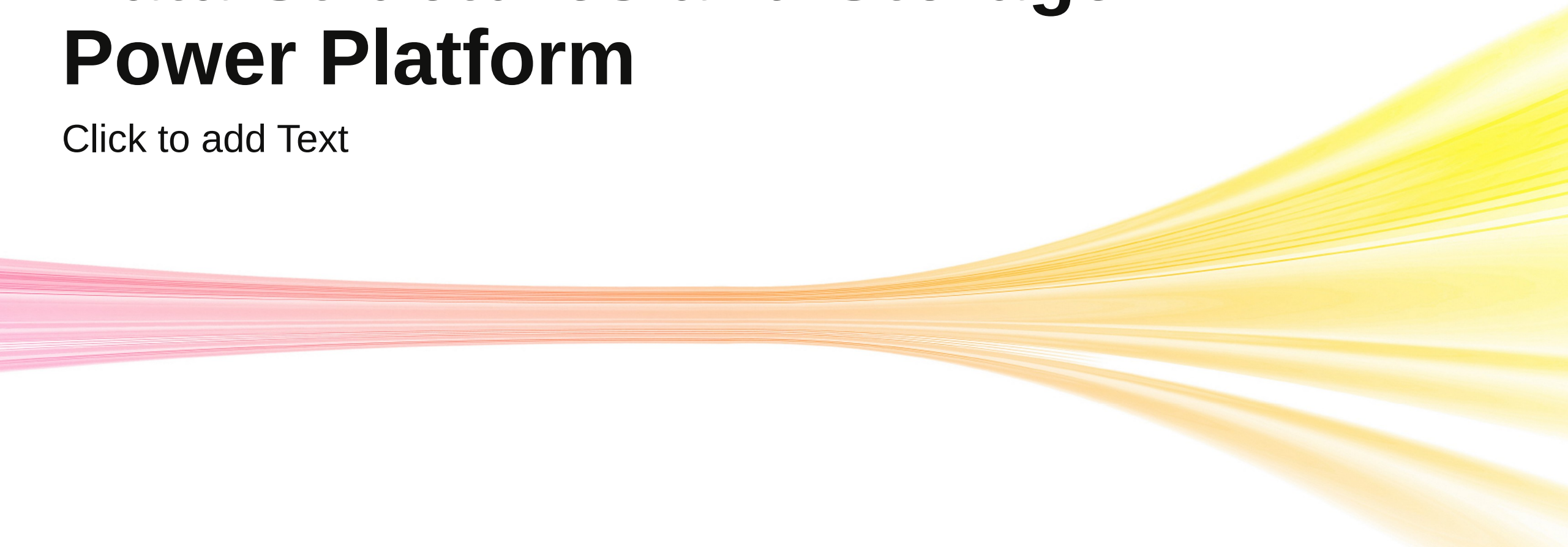
- Analyse different storage approaches

Evaluate

- Evaluate suitability for different scenarios

Data Structures and Storage in Power Platform

Click to add Text

A decorative graphic consisting of multiple thin, overlapping wavy lines that create a sense of motion. The lines are colored in a gradient from light pink on the left to bright yellow on the right, with a thin orange line in the middle. They originate from the bottom left and fan out towards the right side of the slide.

Collections vs. Persistent Storage

- Collections exist only in-memory - data is lost when the app is closed
- SharePoint Lists and Dataverse persist data between sessions and users

Dataverse vs. SharePoint Lists

SharePoint Lists vs. Microsoft Dataverse

- **SharePoint Lists:** Excellent for lightweight internal tools. It is included in standard M365 packaging but lacks advanced row-level transactional performance or relational integrity
- **Microsoft Dataverse:** An enterprise-grade, relational cloud database engine. It supports complex data organizations (one-to-many, many-to-many lookups), robust security roles, and deep scalability
- **The Scalability Dividend:** Poorly structured data models limit search efficiency and drop application performance as rows scale past delegation limits

Connecting to a Data Source

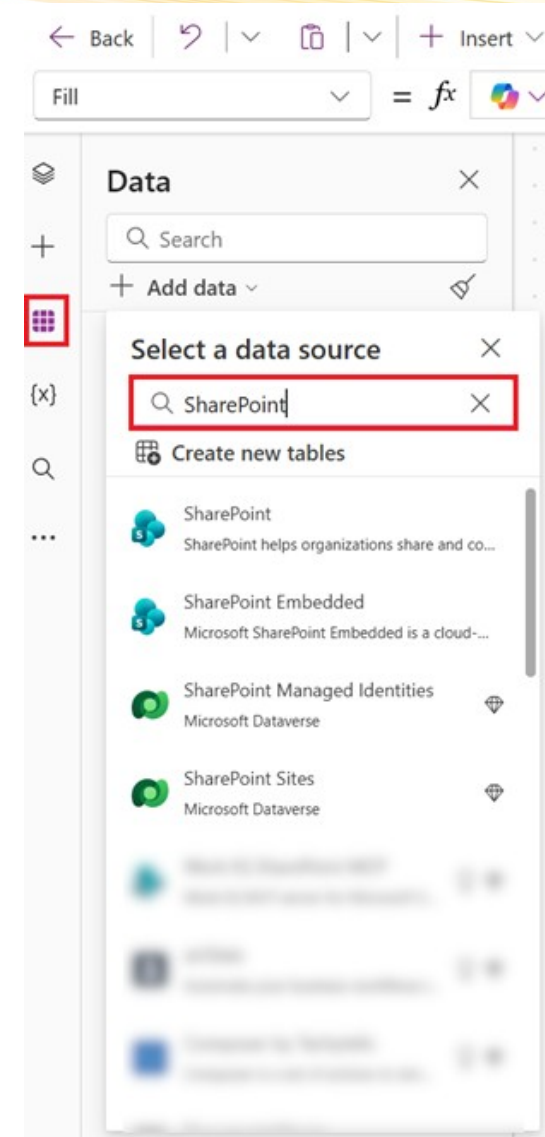
Once connected, a gallery can be bound directly to the data source

Code Example:

```
// Gallery.Items property  
SortByColumns(Products, "Name", Ascending)
```

Connecting & Displaying Backend Data

- **Establish Connection:** Open the left-hand navigation bar, click the **Data** icon (cylinder cylinder), click **Add Data**, search for your target source (SharePoint or Dataverse), and select your table/list
- **The Gallery Binding:** Insert a blank vertical Gallery control onto your canvas screen layout
- **Set the Items Property:** Assign the name of your external data source directly into the Gallery's Items property
- **Mapping Fields:** Select the gallery container, navigate to the properties panel on the right, click **Edit Fields**, and map your UI labels directly to the database column names (e.g., ThisItem.Title)



Submitting and Modifying Records

- **Insert the Edit Form:** Add an **Edit Form control** onto your canvas space and bind its main DataSource property directly to your connected external table
- **Bind Gallery Selection (For Editing):** To make the form display and modify existing records, change the Form's Item property to match your gallery's active row pointer:
Gal_Inventory.Selected
- **Deploy the Action Button:** Add a standard button element to act as the submission trigger
- **Write the Serialisation Formula:** Paste the native Power Fx execution script onto the button's OnSelect event property

// Submit form state and automatically reset the visual inputs

SubmitForm(Frm_ProductDetails);

NewForm(Frm_ProductDetails);

Notify("Cloud database successfully updated!",
NotificationType.Success)

Populating the Form

Connecting the Gallery to Your Form:

The Technical Challenge

- By default, an **Edit Form** control sits completely blank because it has no idea *which* record in your database you want to modify
- You must programmatically pass the selected row from your UI visual list directly into the form container's memory asset

Step-by-Step Implementation Rule:

- Select your **Form control container** on the left-hand tree view
- In the properties drop-down menu on the top-left of the formula bar, select the **Item** property
- Paste the following explicit Power Fx locator formula into the bar:

[YourGalleryName].Selected

Technical Data Flow Verification

The Verification Checklist:

- **Run the App:** Hold down the **Alt key** (or click the Play icon) to interact with your live form layout interface
- **Submit Mock Entry:** Type a distinct test string into the text inputs and press your submit button trigger
- **Audit the Backend:** Open your host database (SharePoint list viewer or Dataverse table designer tab) in your browser and refresh the page to confirm the row exists
- **Confirm the UI Loop:** Check your connected application gallery. It should automatically display the fresh data row without requiring an app reload

Handling Asynchronous Caching

The Technical Challenge:

- External cloud data connectors (like SharePoint and Dataverse) use a performance caching layer to keep mobile applications fast
- **The Bug:** When a user updates a backend database table externally or programmatically via a custom script, the application UI gallery may refuse to redraw because it is reading stale data trapped in its local memory cache

The Solution: Forcing a Server Pull:

- To resolve UI de-synchronisations, you must explicitly instruct the platform to clear its internal memory cache layer and fetch a fresh row set directly from the server
- **The Command:** Use the Refresh() function immediately following your data submission

Code Implementation

// Execute programmatic update and clear the caching layer

Patch(InventoryDataSource, Defaults(InventoryDataSource), { ProductName: Txt_Entry.Text });

Refresh(InventoryDataSource); // Purges local cache and pulls fresh server rows

Notify("Item successfully committed and synced!", NotificationType.Success)

Adding and Updating Records

Patch() writes a new or updated record to a connected data source

Code Example:

```
Patch(  
    Products,  
    Defaults(Products),  
    { Name: NameInput.Text, Price: Value(PriceInput.Text) }  
)
```

Choosing the Right Storage

Poor Choice - Collection for Business Data:

// Ongoing stock data kept only in a collection

`ClearCollect(colStock, ...)` // lost when app closes!

Better Choice - Dataverse:

// Gallery.Items bound directly to Dataverse table

`SortByColumns(Products, "Name", Ascending)`

Why is a collection unsuitable for data that must persist across sessions?

- Collections store data exclusively inside a device's temporary volatile memory (RAM)



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Hands-On Exploration

Hands-on Data Connection Activity



Structured Practice with Data Sources:

- Connect a Power App to an external data source
- Display records using galleries
- Create forms to add, edit and submit records

Tasks:

- Update records and verify changes within the data source
- Observe how data structure influences UI behaviour and app logic

• **Expected Outcome:** Working Power Apps solutions demonstrating successful connection, display and management of external data sources

Scenario-Based Decision Activity



ACTIVITY



Workplace-Style Scenarios (shown on next slide):

- Temporary data collection
- Ongoing business data
- Reporting and analytics datasets

Tasks:

- Select the most appropriate storage solution for each scenario
- Justify the choice based on persistence, performance and usability
- Identify potential limitations of the selected option

Expected Outcome: Clear reasoning demonstrating an understanding of when to use Collections, SharePoint Lists or Dataverse



ACTIVITY



Workplace Prompts

- **Case Prompt A:** A fast-deploying field inspection app tracking vehicle checkouts for a localised department of 25 staff
- **Case Prompt B:** A real-time data capturing grid recording quick user validation bugs during a 1-day live training event
- **Case Prompt C:** A corporate asset ledger tracking multi-currency inventory lines across global sites, requiring granular row-level data restrictions and relational foreign-key linking

Inventory Management App



ACTIVITY



Objective: Build an app that stores product data in a structured data source

Tasks / Requirements:

- Store product data within an appropriate structured data source
- Display records using galleries
- Implement add and edit functionality through forms

Key Point: Justify the storage choice based on persistence, performance and scalability

Extension Activity: Add sorting options and additional product attributes



ACTIVITY



The Column Reference Quirk

The Technical Quirk

- Power Fx uses two different functions for sorting data: `Sort()` and `SortByColumns()`
- While they achieve similar outcomes, they handle column names completely differently
- Mixing up their syntax rules will trigger immediate compilation failures

The Explicit Syntax Rules

- Approach 1: Using **`Sort()`**
- Approach 2: Using **`SortByColumns()`**

Approach 1: Using Sort()

Rule: Pass the column name as a raw object reference (no quotes).

Syntax blueprint:

```
Sort(InventoryDataSource, Price, SortOrder.Descending)
```

Approach 2: Using SortByColumns()

Rule: You **must** wrap the target column name inside text literal double quotation marks.

Syntax blueprint

`SortByColumns`(InventoryDataSource, "Price", SortOrder.Descending)

TIP: If a sorting formula throws a red wavy underline error, check your quotation marks!

Debugging and Refinement Activity

Common Issues Provided:

- **Fault A:** Caching layer issue
- **Fault B:** Schema casing/naming issue
- **Fault C:** Authentication layer issue
- **Fault D:** Security/Governance issue

Tasks:

- Identify and annotate the cause of the issue
- Correct configuration or formula errors
- Test the solution to confirm resolution
- Verify data integrity across the application

Expected Outcome: Evidence of a structured troubleshooting approach that considers data sources, connections, permissions and UI behaviour



ACTIVITY



ACTIVITY



Evaluation Discussion

Learners discuss and compare:

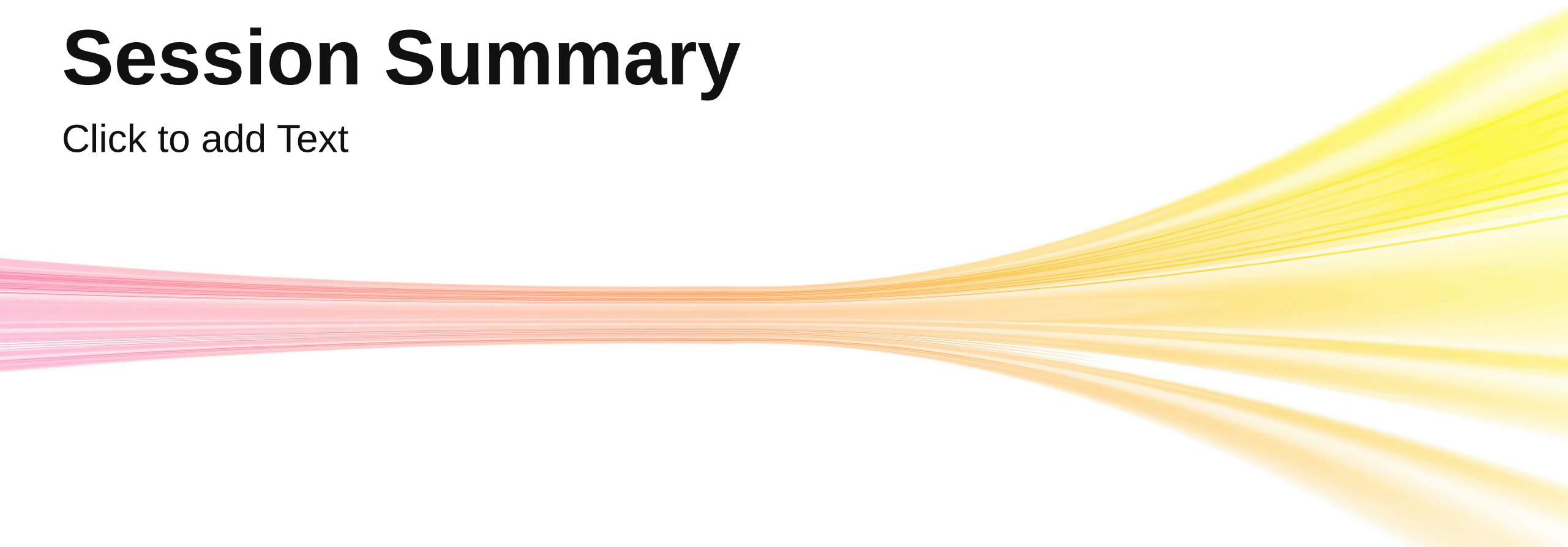
- Scalability
 - Data structure and organisation
 - Performance considerations
 - Business suitability
 - Maintainability
-
- **Expected Outcome:** Learners can explain the advantages, limitations and appropriate use cases for Collections, SharePoint Lists and Dataverse

Stretch and Challenge

- Create related datasets, e.g. product categories linked to products
- Implement relationships between records
- Justify data structure and schema design decisions

Session Summary

Click to add Text



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